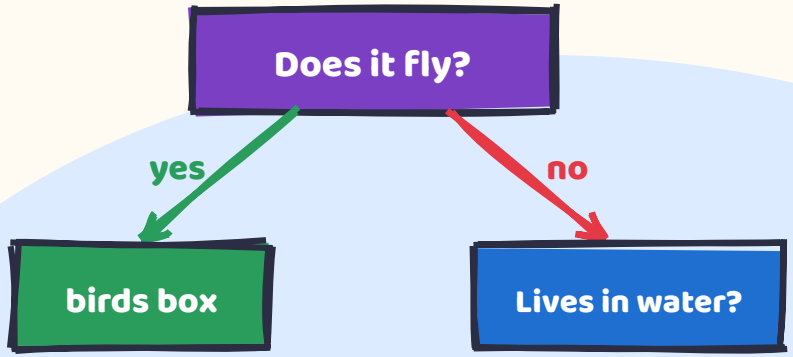
Little Explorers · Class 16

Robot Sorting Rules Game

Teach a robot to sort, just like real AI!

Yes / no questions | Question tree | Fixing rules



Did you know?
Some recycling machines use AI to sort plastic, paper and metal. They follow rules learned from thousands of examples!

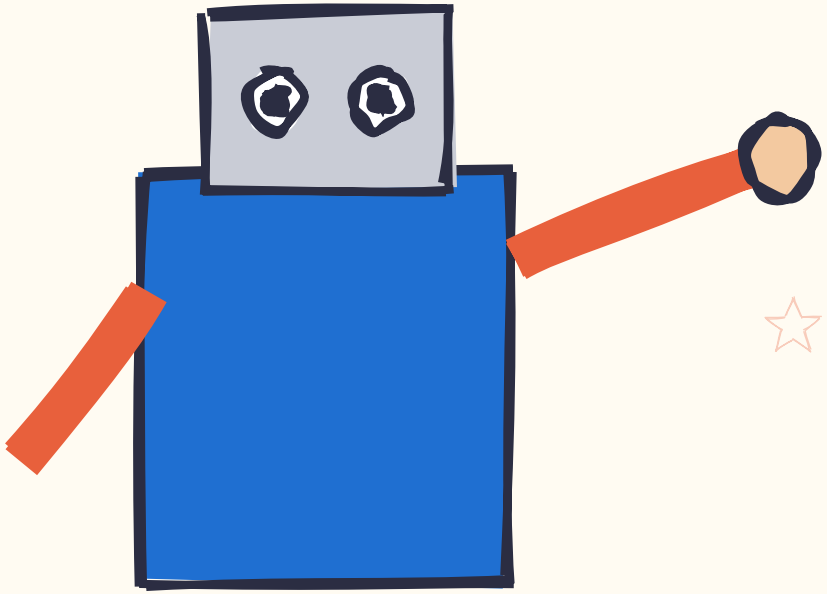
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Let's wonder...

How does a computer sort things?

Too many pictures!

Your phone sorts photos into **animals**, **food** and **people** all by itself. How does it decide? Can we make rules that a robot could follow?



It guesses?

It looks closely?

It follows rules!

Today: we play a sorting game where **you** write the rules and a friend plays the **robot**!

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Learning objectives

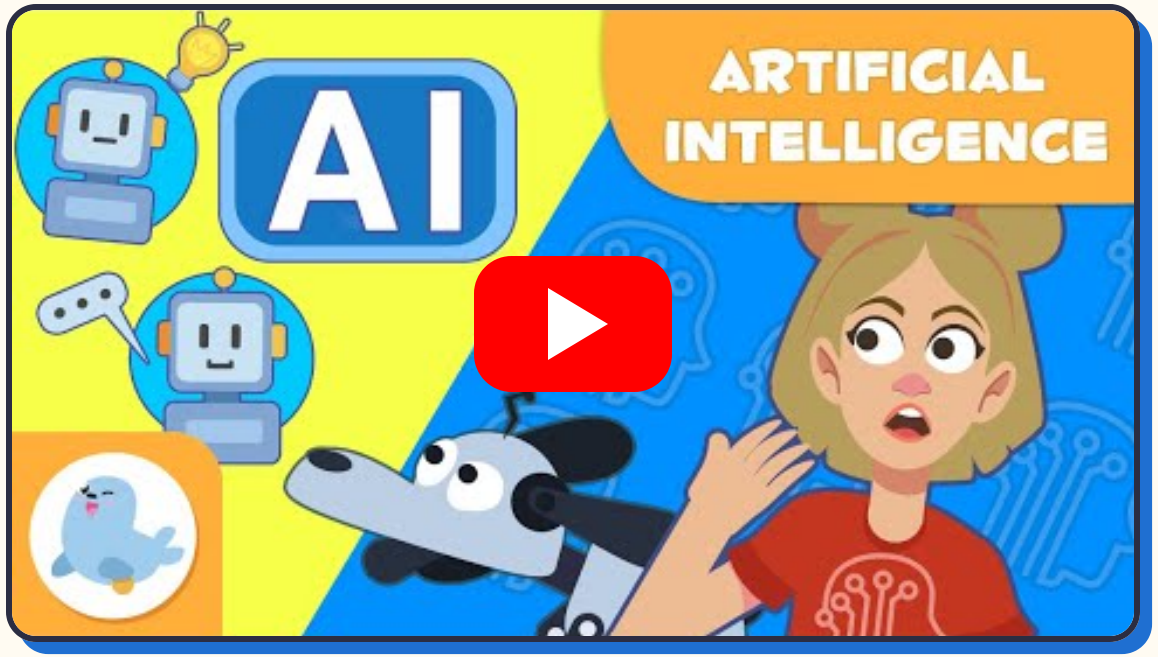
Today I can...

1	**Yes or no:** sort things using **yes / no questions**	**Proof:** we sorted all our picture cards
2	**Question tree:** draw my rules as a **question tree**	**Proof:** my tree has at least 2 questions
3	**Fix the rules:** fix my rules when the **robot** makes a mistake	**Proof:** I changed a rule after a mistake

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Objective 1 · Yes or no · video

Video time!



Artificial Intelligence for Kids

Smile and Learn · about 3 minutes · youtu.be/G6de8L7cVvM

Watch for...

1. What is **artificial intelligence** (AI)?
2. How does AI **learn**?
3. Name one thing AI can help us with.

Scan to watch

After the video: Tell your partner one place you have seen AI (a phone, a speaker, a game?).

Objective 1 · Yes or no

Sort with yes or no

The idea

A good sorting question has only two answers: **yes** or **no**.

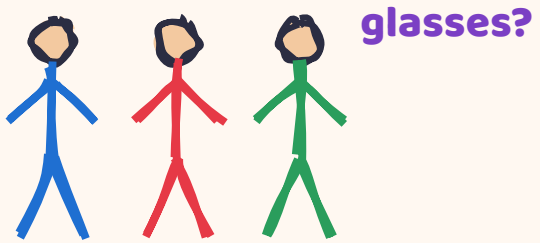
"Does it fly?" Yes: bird, plane, butterfly. No: fish, cat, car.

With just a few yes/no questions we can sort **lots** of things!

Careful

'What colour is it?' is NOT a yes/no question. 'Is it red?' IS!

Example: Guess Who?



In the game 'Guess Who?' you ask: "Does your person wear glasses?" Yes or no? Each answer removes lots of faces until only one is left!

Objective 2 · Question tree

Draw a question tree

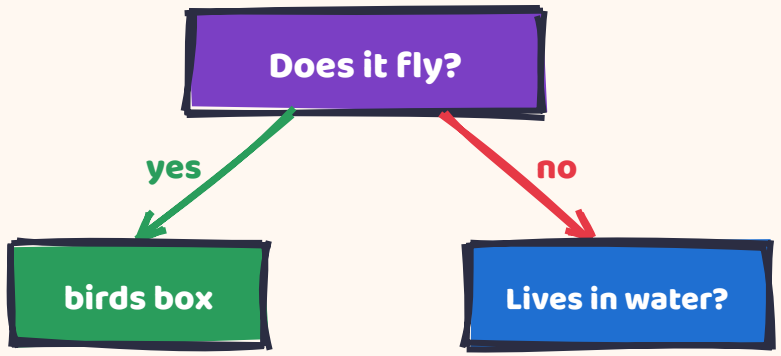
The idea

Put your first question at the top. Draw a **yes** branch and a **no** branch. Then ask a **new question** on each branch.

This is a **question tree** (computers call it a **decision tree**).

Big trees
AI that sorts photos and recycling uses trees with thousands of questions!

Example: a real tree



A tree trunk splits into branches, and each branch splits again into smaller branches. Our questions split our cards the same way!

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Objective 3 · Fix the rules

Fix the rules, not the robot

The idea

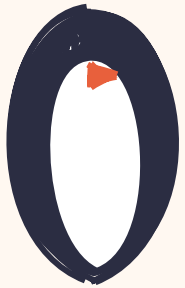
If a card goes in the wrong group, look at your tree. Which question was **confusing**?

Make the question **clearer** or **add a new question**. Then test again with the robot!

Real AI

Real AI makes mistakes too. Engineers keep testing and improving it.

Example: a penguin card



Our rule said: 'Is it a bird? Then it flies.' But a **penguin** is a bird that cannot fly! We fix the rule: ask 'Can it fly?' instead of 'Is it a bird?'

LEARN

Real world

Machines that sort



Recycling
Sort plastic, paper and metal



Photos
Group pictures of pets, food and friends



Post offices
Sort letters by their address



Farms
Sort good fruit from bruised fruit



What would you like a sorting robot to sort at home?

Clean data
AI learns from examples. Wrong examples teach it wrong things!

Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

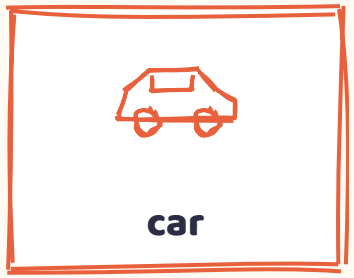
1. Sort the picture cards
2. Be the robot!
3. Robot test results
4. Sorting challenges

Objective 1 · Yes or no · sort the cards

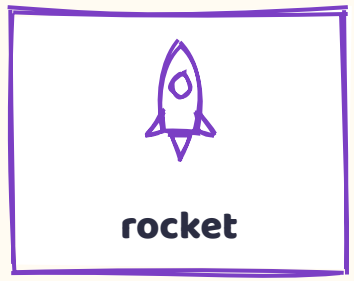
Sort the picture cards



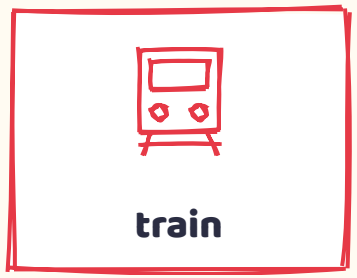










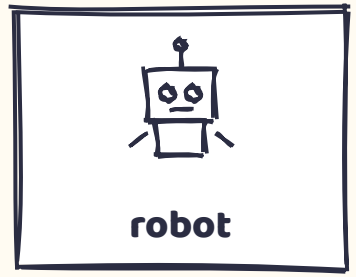












Your job

Think of a yes/no question that splits the cards into **two** groups. Try: **Can it fly?** or **Does it have wheels?**

Any rule!
There is no single right answer. Different teams make different rules!

Class game

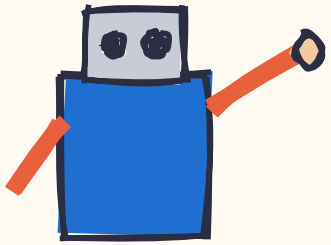
Be the robot!

How to play

1. One child is the **robot**. Robots cannot think: they only follow the rules.
2. Give the robot your question tree.
3. Hand the robot **new** picture cards it has not seen.
4. The robot follows the tree **exactly** and sorts each card.
5. Did any card go in the wrong box?

Then discuss

When the robot made a mistake, was it the **robot's** fault or the **rules'** fault?



Big idea

A machine is only as smart as the rules and examples we give it.

MAKE

Test the robot!

Robot test results

New card	Robot put it in…	Correct?
Card 1: _____	_____	right / wrong
Card 2: _____	_____	right / wrong
Card 3: _____	_____	right / wrong
Card 4: _____	_____	right / wrong

Talk about it

How many did the robot get right? What rule will you **change**?

Fix and retry

Change one question, then test the same cards again.

MAKE

Level up!

Sorting challenges

Choose your challenge



Three groups: make a tree that sorts cards into 3 groups.



Mystery card: the robot gets a card nobody has seen. Does your tree work?



Swap trees: use another team's tree. Is it clearer or harder?



Fewest questions: sort all the cards with as few questions as possible.

Think like an engineer

Why might a robot sort a toy car and a real car into the same group?

Boss level

Make a tree that sorts every card into its own group!

Think about it

Are our rules fair?

Sorting people?

Imagine a robot sorting **children** by a rule like 'Is your hair long?' to choose a team. Is that fair? What would be a **fairer** rule?

Good rules are…

- ✔ clear (everyone understands)
- ✔ fair (no one is left out unfairly)
- ✔ tested (we checked them)

Big idea

People write the rules for AI. So people must make sure the rules are kind and fair.

Objective check

Can I really do it? Prove it.

01 Say one good yes/no sorting question.

02 Show your question tree. How many questions does it have?

03 What mistake did the robot make, and how did you fix your rules?

Rate yourself

Colour one circle each. Red: not yet. Amber: with help. Green: I can explain it to a friend.

Teacher: Display all the question trees on the wall. Celebrate the different ways to sort!

Show what you know

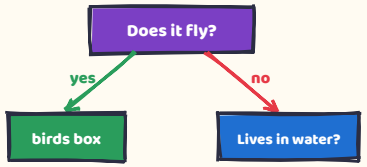
What did we learn?

- [x] Yes/no questions can sort lots of things
- [x] A question tree puts the questions in order
- [x] Robots only follow rules: they do not think
- [x] When the robot is wrong, fix the rules
- [x] Good rules are clear and fair

Exit ticket: why did the penguin card go in the wrong group?

At home

Sort your toys or clothes with a yes/no question tree!



Next class:
Phase 2: more robots and coding!